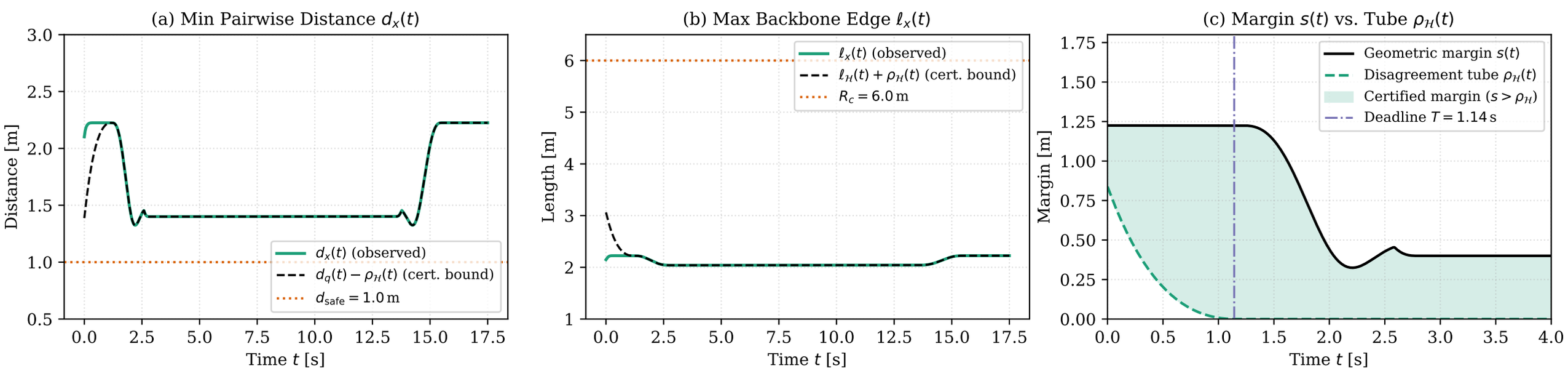
(a) Min Pairwise Distance $d_x(t)$
Distance [m]
Time $t$ [s]
$d_x(t)$ (observed)
$d_q(t) - \rho_{\mathcal{H}}(t)$ (cert. bound)
$d_{\text{safe}} = 1.0\,\text{m}$
(b) Max Backbone Edge $\ell_x(t)$
Length [m]
Time $t$ [s]
$\ell_x(t)$ (observed)
$\ell_{\mathcal{H}}(t) + \rho_{\mathcal{H}}(t)$ (cert. bound)
$R_c = 6.0\,\text{m}$
(c) Margin $s(t)$ vs. Tube $\rho_{\mathcal{H}}(t)$
Margin [m]
Time $t$ [s]
Geometric margin $s(t)$
Disagreement tube $\rho_{\mathcal{H}}(t)$
Certified margin ($s > \rho_{\mathcal{H}}$)
Deadline $T = 1.14\,\text{s}$